eg: $F_{drag}=-\beta v$ $\longleftarrow\boxed{}\longrightarrow$ F_{prop}

1) $F_{prop}=F_{drag}=\beta v \Rightarrow v=\frac{F_{prop}}{\beta}$ (const velocity).

2) $v(0)=\frac{F_{prop}}{\beta}$

$\frac{1}{4}F_{prop}-F_{drag}=ma$

$\frac{1}{4}F_{prop}-\beta v=m\frac{dv}{dt}$

$\int_{v(0)}^{v(t)}\frac{dv}{\frac{1}{4}F_{prop}-\beta v}=\int_0^t\frac{1}{m}dt$

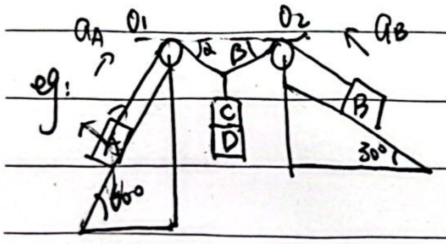
$-\frac{1}{\beta}\ln(\frac{1}{4}F_{prop}-\beta v)\Big|_{v(0)}^{v(t)}=\frac{1}{m}t$

$\frac{\frac{1}{4}F_{prop}-\beta v}{\frac{1}{4}F_{prop}-\beta v_0}=e^{-\frac{\beta}{m}t}$

$\frac{4\beta v-F_{prop}}{3F_{prop}}=e^{-\frac{\beta}{m}t}$

$v(t)=\frac{F_{prop}}{4\beta}(3e^{-\frac{\beta t}{m}}+1)$.

eg:

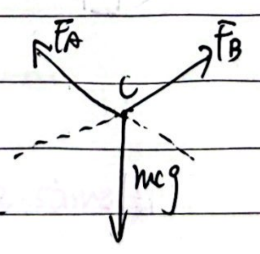


A: $T_A-m_Ag\sin60°=0\Rightarrow T_A=mg$

B: $T_B-m_Bg\sin30°=0\Rightarrow T_B=mg$

$\begin{cases}T_A\cos\alpha=T_B\cos\beta\\T_A\sin\alpha+T_B\sin\beta=mg\end{cases}\Rightarrow \alpha=\beta=30°$

adding D:



A: $F_A-m_Ag\sin60°=m_Aa_A$

B: $F_B-m_Bg\sin30°=m_Ba_B$

point C: $m_Cg\sin30°+F_A\sin30°-F_B=m_C\cdot a_B$

$m_Cg\sin30°+F_B\sin30°-F_A=m_C\cdot a_A$ $\Rightarrow a_A=\frac{15g}{48+14\sqrt{3}}$ $a_B=\frac{2+\sqrt{3}}{5}a_A=\frac{3(2+\sqrt{3})}{48+14\sqrt{3}}g$

$m_C=m+m=2m$

Mid RC

1) $f = -kv$ $\qquad f = kv \leftarrow \square$

$f = m\frac{dv}{dt} = -kv$

$\frac{dv}{dt} = -\frac{kv}{m}$

$\frac{dv}{v} = -\frac{k}{m}dt$

$\int_{v_0}^{v} \frac{1}{v} dv = -\frac{k}{m}\int_0^t dt$

$\ln\frac{v}{v_0} = -\frac{kt}{m}$

$v(t) = v_0 e^{-\frac{k}{m}t}$

$L(t) = \int_0^t v(t)dt = \int_0^t v_0 e^{-\frac{k}{m}t}dt = \frac{m}{k}v_0(1-e^{-\frac{k}{m}t})$

2) $f = -kv$, free fall $\qquad \uparrow f = kv \quad \downarrow mg$

$m\frac{dv}{dt} = mg - kv$

$\frac{dv}{mg-kv} = \frac{dt}{m}$

$\int_0^v \frac{1}{mg-kv}dv = \frac{1}{m}\int_0^t dt$

$-\frac{1}{k}\ln(mg-kv)\Big|^v = \frac{1}{m}t$

$\ln\frac{mg-kv}{mg} = -\frac{k}{m}t$

$v(t) = \frac{mg}{k}(1-e^{-\frac{k}{m}t})$

$L(t) = \frac{mg}{k}\int_0^t(1-e^{-\frac{kt}{m}})dt = \frac{mg}{k}[t - \frac{m}{k}(1-e^{-\frac{k}{m}t})$

3) $f = -kv^2$. $\qquad f = -kv^2 \quad \leftarrow \square$

$m\frac{dv}{dt} = -kv^2$

$\frac{dv}{dt} = -\frac{k}{m}v^2$

$\frac{1}{v^2}dv = -\frac{k}{m}dt$

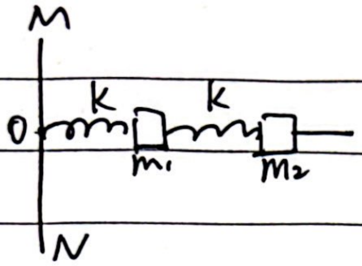
$\int_{v_0}^{v}\frac{1}{v^2}dv = -\frac{k}{m}\int_0^t dt$

$\frac{1}{v} - \frac{1}{v_0} = \frac{k}{m}t$

$v(t) = \frac{1}{\frac{1}{v_0}+\frac{k}{m}t}$

$L(t) = \int_0^t v(t)dt = \int_0^t \frac{1}{\frac{1}{v_0}+\frac{k}{m}t}dt = \frac{m}{k}\ln(1+\frac{kv_0}{m}t)$

eg:



original l_0, l_0, now: left: $l_1 > l_0$, right: $l_2 > l_0$

m_1: $k(l_1-l_0)-k(l_2-l_0)=m\omega^2 l_1$ ①

m_2: $k(l_2-l_0)=m\omega^2(l_1+l_2)$ ②

Let $A=\frac{m\omega^2}{k}$, rewrite ① & ②

$\therefore l_1-l_2=Al_1 \Rightarrow l_1(1-A)=l_2$ $\Rightarrow$ $l_1=\frac{1}{A^2-3A+1}l_0$

$l_2-l_0=A(l_1+l_2)$ $\quad l_2=\frac{1-A}{A^2-3A+1}l_0$

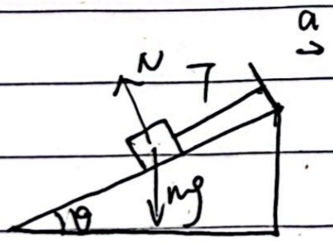
Since $l_1>l_0$, $l_2>l_0$

$0<A^2-3A+1<1 \Rightarrow 0<A<3,\ A<\frac{1}{2}(3-\sqrt{5}),\ A>\frac{1}{2}(3+\sqrt{5})$ $\Big\}$ $0<A<\frac{1}{2}(3-\sqrt{5})$

$\frac{1-A}{A^2-3A+1}>1 \Rightarrow 0<A<2$

$\omega<\frac{1}{2}(\sqrt{5}-1)\sqrt{\frac{k}{m}}$.

eg:



$T-mg\sin\theta=ma\cos\theta$

$mg\cos\theta-N=ma\sin\theta$

$T<2mg \Rightarrow 2mg>mg\sin\theta+ma\cos\theta$ $\quad a<\frac{2-\sin\theta}{\cos\theta}g$

$N\geq0$. $g\cos\theta-a\sin\theta\geq0$ $\quad a\leq g\cot\theta$

compare $\frac{2-\sin\theta}{\cos\theta}$ and $\cot\theta$. when $\frac{2-\sin\theta}{\cos\theta}=\cot\theta=\frac{\cos\theta}{\sin\theta} \Rightarrow \theta=\frac{\pi}{6}$

when $\theta\geq\frac{\pi}{6}$, $\frac{2-\sin\theta}{\cos\theta}\geq\cot\theta$, when $\theta<\frac{\pi}{6}$, $\frac{2-\sin\theta}{\cos\theta}<\cot\theta$

when $\theta\geq\frac{\pi}{6}$, $a\leq g\cot\theta$

when $\theta<\frac{\pi}{6}$, $a<\frac{2-\sin\theta}{\cos\theta}g$